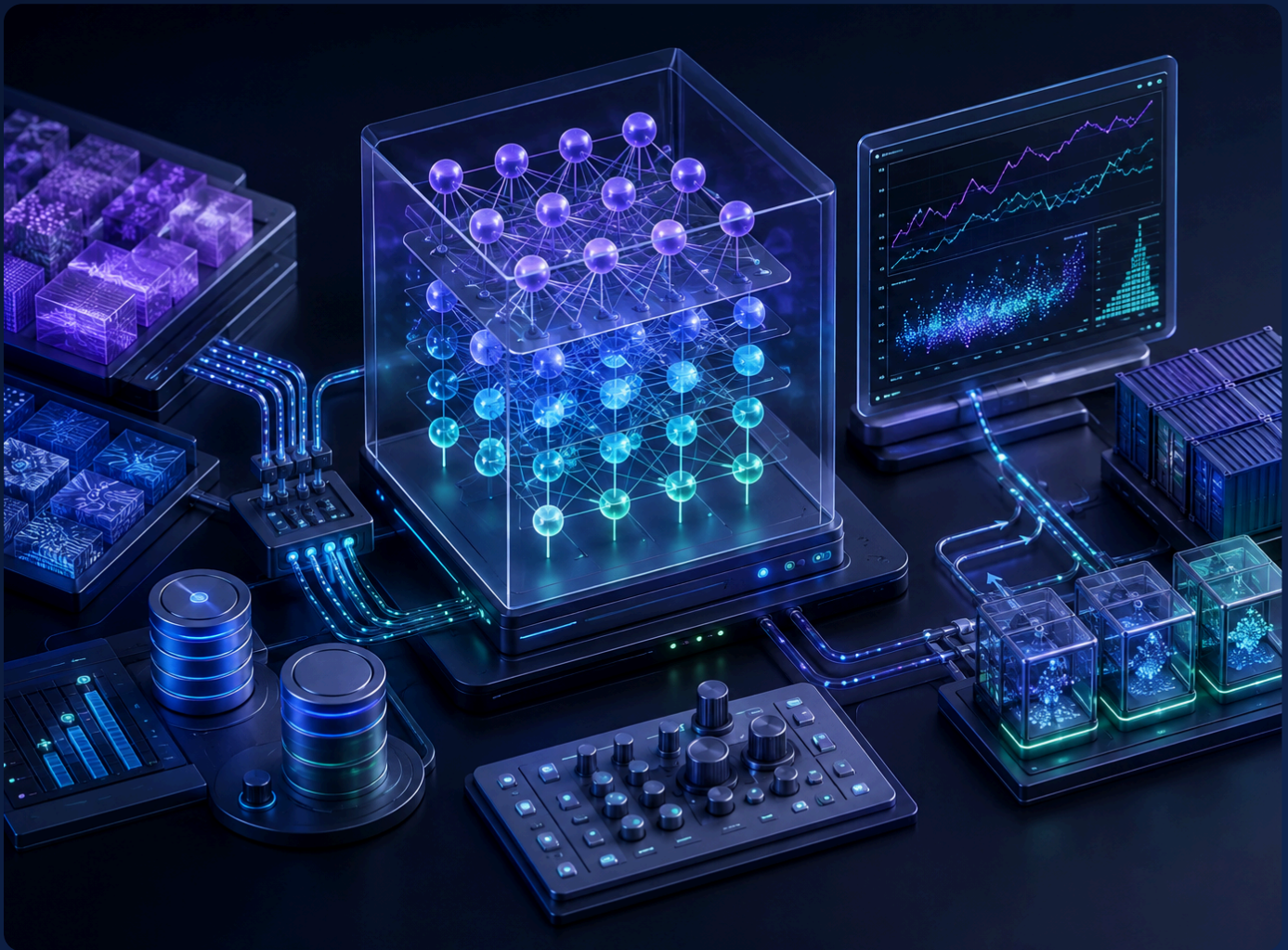


ARTIFICIAL INTELLIGENCE · MULTI-LEVEL

Applied Machine Learning Engineer Path

Build models. Test assumptions. Deliver useful ML.



Data → ML → Deep Learning → CV/NLP → LLM/SLM Engineering → MLOps.

🕒 20-28 weeks

📁 6 modules

📖 24 lessons

[Explore the course →](#)

Concepts, worked examples, assignments and a coherent course project.

WHO IT IS FOR · WHAT YOU WILL LEARN

Your next capability.

A production-oriented machine learning path spanning classical ML, deep learning, computer vision, NLP, voice, multimodal AI, model adaptation and MLOps.

 Build and evaluate classical ML systems Develop deep learning solutions for CV and NLP Fine-tune models with PEFT, LoRA and QLoRA Train a small educational transformer Deploy and monitor models with MLOps **Audience**

- AI and data students
- Data analysts
- Software developers entering ML
- Career switchers

 Prerequisites

- Basic programming
- Python and math bridges are included

FROM CONCEPT TO EVIDENCE

**One curriculum, three connected views**

This guide summarizes the same modules and lessons shown on the course overview and in the learning workspace. The next pages explain the scope and practical evidence for every module.

DETAILED MODULE CURRICULUM · 1-3

Build the foundations.

The module names, lesson sequence and evidence below match the course learning workspace.

1 Data, Math and Python Bridge

Prepare a reproducible dataset and a defensible prediction question.

- 1.1 Python for data**
Use Python transformations with explicit input and output shapes.
- 1.2 Statistics essentials**
Statistics helps describe variation and uncertainty.
- 1.3 Data preparation**
Define the prediction time before selecting features.
- 1.4 SQL and visualization**
SQL extracts records at a defined grain; visualization reveals distributions and quality problems.

You will produce: Dataset contract and leakage review.

2 Classical Machine Learning

Compare simple models using an evaluation design that reflects deployment.

- 2.1 Regression and classification**
Regression predicts a quantity; classification predicts a category or class probability.
- 2.2 Feature engineering**
Features should encode information available at inference time.
- 2.3 Clustering**
Clustering groups examples according to a representation and similarity choice.
- 2.4 Metrics and model selection**
Select metrics that match class balance and error costs.

You will produce: Model comparison and evaluation memo.

3 Deep Learning

Understand how neural networks learn and where training can fail.

- 3.1 Neural network foundations**
A neural network composes parameterized transformations.
- 3.2 Optimization**
Optimization updates parameters to reduce training loss.
- 3.3 CNNs**
Convolutional layers share weights across spatial positions.
- 3.4 Sequence and transformer foundations**
Sequence models operate on ordered representations; attention relates positions through learned scores.

You will produce: Training notebook, shape diagram and learning curves.

DETAILED MODULE CURRICULUM · 4-6

Apply, integrate and demonstrate.

The module names, lesson sequence and evidence below match the course learning workspace.

4 Computer Vision, NLP and Multimodal

Evaluate perception tasks according to their actual outputs.

4.1 Classification and detection

Classification labels an image; detection locates and labels objects.

4.2 Segmentation, tracking and OCR

Segmentation labels regions, tracking associates observations across time and OCR extracts text.

4.3 NLP and transformers

NLP workflows require a defined text unit, label policy and evaluation set.

4.4 Voice and multimodal workflows

Combining voice, image and text does not remove uncertainty from the inputs.

You will produce: Perception evaluation and error analysis.

5 LLM and SLM Engineering

Adapt a model only after defining the behavior and evaluation gap.

5.1 Dataset preparation and evaluation

Training data needs provenance, consistent formatting and an explicit label policy.

5.2 Supervised fine-tuning

Supervised fine-tuning adjusts a pretrained model using target examples.

5.3 PEFT, LoRA and QLoRA

Parameter-efficient adaptation updates a subset or additional low-rank parameters.

5.4 Quantization and distillation

Quantization reduces representation precision; distillation trains a smaller model to approximate useful behavior.

You will produce: Adaptation experiment and model card.

6 Educational Transformer and MLOps

Connect an educational transformer to a reproducible serving workflow.

6.1 Tokenizer and embeddings

A tokenizer maps input text to token identifiers; embeddings map identifiers to learned vectors.

6.2 Transformer blocks and training loop

A training loop performs forward computation, loss evaluation, gradient computation and parameter updates.

6.3 Model serving

Serving needs a versioned input contract, resource limits and health checks.

6.4 Monitoring, drift and capstone

Monitoring tracks data quality, latency and task outcomes after deployment.

You will produce: Educational model, serving prototype and MLOps report.

PROJECTS AND WORKING METHODS

End-to-end prediction service

Prepare a dataset, establish a baseline, compare models and package a reproducible prediction workflow with documented limitations.



Reproducible data pipeline



Model evaluation report



Prediction service and monitoring plan



Tools and methods

Python and Pandas

scikit-learn

PyTorch

OpenCV and YOLO concepts

Transformers and PEFT

Model serving and MLOps

Tools are learning choices, not partner endorsements.

Inside module 1: a worked example

Prediction: churn in the next month

Allowed feature: activity before the cutoff

Excluded feature: account closure reason populated afterward

Split: earlier periods for training; later held-out period for evaluation

Fit preprocessing only on training data.

A post-outcome field can make a model appear accurate by revealing information unavailable at prediction time.

How your work is reviewed

Show the artifact, explain your decisions, test the stated assumptions and identify limitations. Module assignments build toward the course project rather than replacing it with a single example.

DELIVERY, COMPLETION AND NEXT STEPS

Know what you are signing up for.

**Delivery**

Hybrid learning · 20–28 weeks

Confirm current schedule, cohort arrangements and enrollment terms with the academy.

**Completion requirements**

Complete the required coursework and module assignments, and pass the course assessment.

A course completion certificate is available after those requirements are met.

What the learning workspace includes**Follow the modules**

Expand the curriculum and open the exact lesson you need.

**Apply the concepts**

Read explanations, inspect examples and complete module practice.

**Keep your evidence**

Use resources, notes, discussion and assignments to support your learning.

Learning commitments, not outcome guarantees

Completion is not a placement, employment or performance guarantee. The course project is educational evidence; any production use requires appropriate review and validation.

Start with the curriculum. Continue with the first lesson.

Review your starting point, download this guide and explore the connected course experience.

[Go to the first lesson](#) →

Client-review edition. The static learning workspace saves demonstration progress locally; production enrollment, assessment, communications and certificate issuance are not connected.